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Quantifying the social impacts of the London Night Tube with a double/debiased machine learning based difference-in-differences approach

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ABSTRACT

There is a worldwide trend toward a growing number of people involved in various night-time activities. The night-time public transport service is of central importance for the urban night-time mobility. In London, the Night Tube service was launched in 2016 to meet the constantly growing night-time travel demand and support London's night-time economy. Yet limited empirical evidence on the ex-post impacts of the London Night Tube has been provided. In this study, we conduct a causal analysis on such impacts using a double/debiased machine learning based difference-in-differences approach. Specifically, we quantify the impacts of the Night Tube on London's night-time economy, house prices, road crashes and related casualties, and crimes. We further investigate the spatial variations in such impacts. Our results indicate a rise in house prices associated with the announcement and the implementation of the service. The number of night-time workplaces showed a limited response. Regarding the safety dimension, we find that the Night Tube service led to a small reduction in the frequency of road crashes but a substantial reduction in crash-related casualties. However, the crime rate in areas served by the Night Tube was increased, especially for the following two categories, robbery of personal property and violence against the person. Moreover, the impact on the crime rate is found to be larger in the inner London area. These findings provide practical implications for urban planners and policy makers, and reveal the need for monitoring the social impacts of the Night Tube service from a long-term perspective.

1. Introduction

Nowadays, 24-hour and night-time transport strategies are increasingly popular among city governments worldwide (McArthur et al., 2019). Urban night-time public transport services are receiving more attention, which have been recognized as a mean to improve the night-time mobility and boost the night-time economy. As of early 2019, 26 of the 28 capital cities in the European Union had night-time public transport services (Plyushteva and Boussauw, 2020).

For London, a 24-hour global city that is well known for its openness and celebration of diversity, its night-time public transport services are of vital importance (Greater London Authority, 2017; Mayor of London, 2018). Improvements in night-time transport

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systems have been identified as a key priority for the travelers who go out most at night in London. Since 2016, Transport for London (TfL) has launched the Night Tube that provides Friday and Saturday night-time services on a part of London Underground (since August 2016) and Overground (since December 2017) lines. That is, the services on the selected routes operate continuously at weekends from Friday morning to Sunday night. The Night Tube was introduced to meet the constantly growing transport demand at night in London. With the implementation of the Night Tube, night-time journeys in London were predicted to be cut by an average of 20 min (Transport for London, 2014a). Also, people believe that the Night Tube would 'open up London's night-time economy to a whole host of new opportunities' (Volterra Partners, 2014), and make the city more attractive to international visitors and investors. It was estimated that the Night Tube would lead to a generation of around 2,000 permanent jobs, and the net additional output produced as a result equates to an additional £360 million over 30 years (Transport for London, 2014a; Volterra Partners, 2014). As stated in Greater London Authority (2017), 'the launch of the Night Tube has been a huge leap forward' for 24-hour organization of the city.

On the other hand, the night-time economy has always been a contentious issue for London, especially for central London, receiving many complaints of unacceptable levels of noise, activities, and waste (Roberts and Turner, 2005). Moreover, the expansion of night-time economy has been found to create urban spatial hot spots of reported anti-social behaviour and violence due to more night-time activities (e.g., Bromley and Nelson, 2002; van Liempt et al., 2014; Philpot et al., 2019). Regarding the London Night Tube, Papa (2016) pointed out several potential negative social impacts. First, working night shifts can adversely affect the health and well-being of employees and their families. Second, the quality of life of the people living nearby the Night Tube stations, clubs and bars is likely to fall due to the noise of night-time activities. Nonetheless, house prices are likely to rise due to higher accessibility to public transport services and the expansion of night-time economy. As a result, there will be increased gentrification and more debt.

To date, however, very few ex-post evaluation studies have been conducted to formally quantify the social impacts of night-time public transport services. It remains unclear whether and how the London Night Tube has influenced relevant outcomes. This study contributes to the literature by presenting new empirical evidence on the social impacts of the Night Tube. The outcomes of interest include London's night-time economy, house prices, road crashes and related casualties, and crimes. A double/debiased machine learning based difference-in-differences (DMLDID) approach is adopted to make causal inference on such impacts, which outperforms the traditional DID estimation method particularly when handling high-dimensional potential control variables and treatment effect heterogeneity.

The paper is structured as follows. In section 2, we review the relevant literature and introduce the background of the London Night Tube. Section 3 describes the data collected for our analysis. Section 4 provides the theoretical details on the traditional and double/debiased machine learning based DID estimators. Results and discussion are presented in section 5, followed by conclusions in the final section.

2. Background

2.1. Night-time public transport services

The urban night has long been recognized as a timespace of leisure and freedom, an opportunity to escape from the routines of everyday life (Plyushteva, 2019). Since the last century, large numbers of people have been involved in various types of activities that relate to night-time entertainment and retail provision in cities, and such activities constitute the so-called 'night-time economy' (Melbin, 1978; Shaw, 2010). Many European cities realized that there was increasing public demand for going out and having a good time at night in the 1970 s and 1980 s, hence they started to use cultural policies to revitalize urban nightlife (Bianchini, 1995).

Night-time public transport services play an important role in 24-hour cities, which are expected to improve the night-time mobility and boost the night-time economy. For example, the *Estate romana*, an early practice of night-time city in Rome, was supported by cheap and frequent bus services running late into the night (Bianchini, 1995). According to Plyushteva and Boussauw (2020), 26 of the 28 capital cities in the European Union had night-time public transport services by early 2019, and their operation modes varied significantly. However, only a small number of evaluation studies have been conducted on night-time public transport services. Here we summarize three studies that are closely related to our research. First, Arbex et al. (2021) carried out a before-and-after evaluation on the night-time bus service in the city of São Paulo, Brazil. They concluded that the night-time bus service had led to an increase in accessibility. Moreover, higher average travel distance was observed at late-night hours, and hence residents were able to engage in activities that were not available to them before the service improvement. In addition, Lichtman-Sadot (2019) performed an evaluation analysis on Israeli's night-time buses from the perspective of road safety. The implementation of night buses was found to cause a significant reduction in crashes involving young drivers. Also, casualties resulting from crashes decreased. Similarly, an earlier study by Jackson and Owens (2011) evaluated the impacts of expanding the metro service by three hours at night in Washington DC. They found that overall such changes in metro schedule had little impact on drunk driving, alcohol related fatal crashes, or alcohol related arrests, but the areas with bars that were close to transit stations experienced decreases in drunk driving. Such results indicate notable variations in the impacts of night-time transport services (i.e., treatment effect heterogeneity).

2.2. London night Tube

London was termed as 'a city of night owls' by Mayor of London, Sadiq Khan as Londoners are more likely to stay up late than the rest of the UK population. In London, the late-night Tube ridership had grown at double the rate of daytime trips since 2000, and the demand for night-time activities is likely to keep increasing over the next twenty years (Transport for London, 2014a; Mayor of London, 2018). At the end of 2013, Mayor of London and TfL set out vision for the future of the Tube, in which the plan to introduce the

Night Tube service was included due to such constantly growing night-time travel demand (Mayor of London, 2013). In 2014, the Night Tube was officially announced by the Mayor and London Underground (Transport for London, 2014a). The service was initially scheduled to launch on 12 September 2015. However, due to some strike actions, the Night Tube service finally began operation on the night of Friday 19 August 2016 on Central and Victoria lines (Transport for London, 2016). TfL reported that over 50,000 customer journeys were completed using the service on the first night. Then, the Jubilee, Northern, and Piccadilly lines' night-time services started in October, November, and December 2016, successively. It is worth noting that some particular branches of the Central, Northern, and Piccadilly lines are excluded. Service details including routes and launch dates of the Night Tube are provided in Table 1, and the Night Tube network across London is presented in Fig. 1.

A report prepared by *Volterra Partners for TfL and London First* estimated that adding in wider economic impacts, the benefits would be £3.90 for every pound spent on delivering the Night Tube service (Volterra Partners, 2014). On the second anniversary of the Night Tube, TfL described it as a 'bigger success than predicted'. The press release reported that the travel demand on the Night Tube increased from 7.8 million in 2016/17 to 8.7 million in 2017/18 (12 % in percentage), and these increased number of journeys led to an additional GVA (i.e., Gross Value Added) contribution of £190 million to London's economy (Transport for London, 2018).

Recently, several research articles have evaluated the night-time public transport services in London. For instance, McArthur et al. (2019) assessed the equity issue of the Night Tube. They pointed out that the night-time transport strategies in London had failed to improve the transport equity across spatial and temporal dimensions, as the coverage of London's night-time transport network comprising the Night Tube and night bus routes was incomplete and night-time workers faced severe accessibility barriers. Smeds et al. (2020) found that London's night-time transport policies had poorly accounted for differentiated mobility needs and practices. Liu and Cheng (2020) evaluated the first two Night Tube services on Central and Victoria lines using smart card data and open geodemographics. Particularly, they identified the well-performed Night Tube stations (i.e., stations with high night-time travel demand) for both routes, and their results show that both routes had around half of the stations that were well-performed.

Regarding the potential impacts of the London Night Tube on relevant social outcomes, Papa (2016) made a brief discussion. It was suggested that the night-time journeys in London would be faster and cheaper, and therefore more travelers would able to leave their cars at home, which further means less emissions, less pollution, less noise, and fewer road crashes. On the other hand, for night-time employees, working night shifts can have negative impacts on their health, and the exposure to noise and other nuisances that can adversely influence the quality of life is also a critical concern. Nonetheless, Papa (2016) reckoned that the majority of houses would have higher prices, which could possibly lead to increased gentrification and more debt. Similar points have been delivered in some other news reports (e.g., Surrenden Invest, 2016; Independent, 2016). In particular, Surrenden Invest (2016) reported that the house price had risen due to the Night Tube since its announcement in 2014. Moreover, some research articles have indicated the rise in house prices associated with the expansion of night-time economy (e.g., Olt et al., 2019; Pinke-Sziva et al., 2019).

However, limited empirical evidence has been provided yet on the ex-post causal impacts of the Night Tube service. Therefore, in this study, we seek to present a quantitative analysis on such impacts. Specifically, we focus on the following four outcomes that were probably impacted by the service, namely night-time economy, house prices, road crashes and related casualties, and crimes. Detailed discussions on these outcomes will be given in the next section. The findings of this study are expected to enhance the understanding to the bright and dark sides of the Night Tube service and better direct its further improvements.

3. Data

3.1. Outcome variables and covariates

In this study, the outcomes of interest cover economic and safety dimensions. First, a major objective of the Night Tube service is to support London's night-time economy (Volterra Partners, 2014), and thus an ex-post evaluation on the response of the night-time economy is needed. The outcome variable used to measure such response is the number of night-time workplaces, which is included in the night-time economy dataset published by the Office for National Statistics (London Datastore, 2019). The data is provided at borough and middle-layer-super-output-area (MSOA) levels, of which the latter is used for our causal modeling. Note that super-output-areas (SOAs) are geographic areas designed to improve the reporting of small area statistics in England and Wales. There are now 32,844 lower-layer-super-output-areas (LSOAs) and 6,791 MSOAs in England, and 4,835 LSOAs and 983 MSOAs in Greater London (Office for National Statistics, 2021). The London Borough is a higher-level geographic area. Greater London is made up of 32 London boroughs and the City of London. The following four categories of night-time businesses are included in the dataset: (1) cultural and leisure activities, (2) activities which support night time cultural and leisure activities, e.g., retail and urban transport, (3) 24-hour health and personal social services, e.g., hospitals, and (4) activities which support wider social and economic activities, e.g.,

Table 1
Night Tube service details (Transport for London, 2016).

Lines	Routes	Service launch dates
Central line (partial)	Ealing Broadway to Loughton/Hainault	19 August 2016
Victoria line	Walthamstow to Brixton	19 August 2016
Jubilee line	Stratford to Stanmore	7 October 2016
Northern line (partial)	Edgware/High Barnet to Morden via Charing Cross	18 November 2016
Piccadilly line (partial)	Cockfosters to Heathrow Terminal 5	16 December 2016

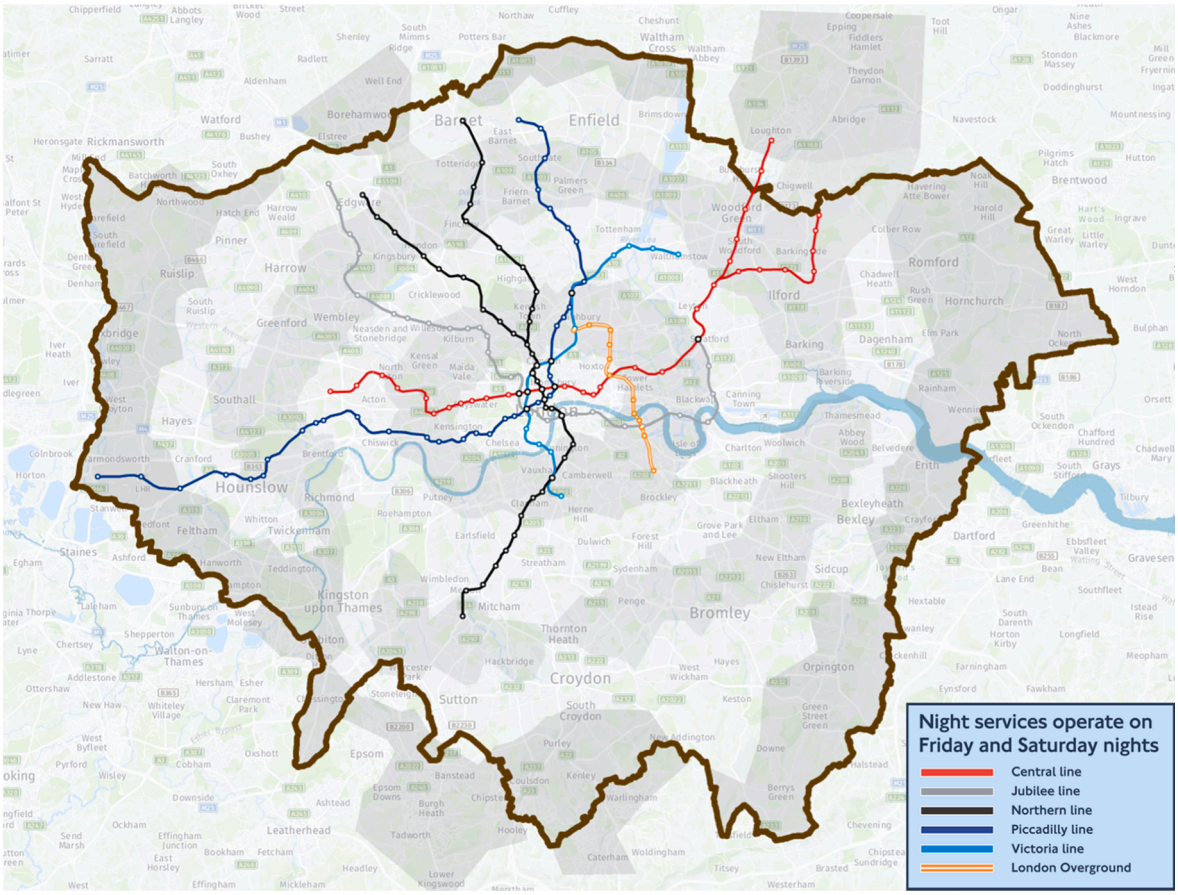


Fig. 1. London Night Tube network (*source*: https://tubecreature.com). *Note*: Grey backgrounds are used to make London fare zones distinguishable.

manufacturing and distribution activities. For more details on night-time industries, please refer to *UK Standard Industrial Classification of Economic Activities 2007* (National Archives, 2016) and *London at night: An evidence base for a 24-hour city* (Mayor of London, 2018). Note that due to the lack of cross-sectional variation, 24-hour health and personal social services are not included in our analysis.

Second, numerous studies have demonstrated that transport infrastructures and policies can either positively or negatively affect house prices (e.g., Efthymiou and Antoniou, 2013; Percoco, 2014; Zhou and Zhang, 2021; Churchill et al., 2021). Moreover, as mentioned previously, Papa (2016) reckoned that even though some houses would be devalued because of exposure to noise and other nuisances, majority of houses would see their values increase in areas served by the Night Tube, which could be due to higher accessibility to public transport services as well as the growth of local night-time economy. Also, Olt et al. (2019) and Pinke-Sziva et al. (2019) have provided empirical evidence on the rising house price associated with the night-time economy. In this study, LSOA level median house price data collected from Land Registry covers the period up to 2018 (London Datastore, 2018). Note that we formalize the impact on the house price in an additive form (Rosenbaum, 2010), and therefore the impact is interpreted as how many more (or less) GBP buyers would pay due to the Night Tube service.

Safety issue is another critical concern for night-time cities (Mayor of London, 2017). We focus on two outcomes, namely road crashes and crimes, to evaluate the Night Tube service from the safety perspective. As mentioned before, Lichtman-Sadot (2019) found that the introduction of late-night buses in Israel decreased the road crashes and related casualties. Similarly, Jackson and Owens (2011) found that the expansion of metro's operation hours at night in Washington DC decreased drunk driving behaviors in some particular areas. Similarly, it is reasonable to assume that the London Night Tube can also reduce night-time road crashes and related casualties by encouraging more travelers to leave their cars at home. The road crash data used in this study is collected from Great Britain's Stats19 database (Department for Transport, 2020). The Stats19 dataset contains three separate files: crash data, vehicle data, and casualty data, which can be integrated by a unique reference number. Because the occurring time of road crashes are available, the analysis of road crashes can be confined to weekend night (i.e., 6 pm Friday – 6am Saturday and 6 pm Saturday – 6am Sunday), which makes it easier to capture the impacts of the Night Tube. We use the period between 6 pm and 6am because it has been used to define 'night' in Mayor of London (2018) and some other official documents. We calculate the crash and casualty rate per thousand population for each LSOA as our outcome variables.

As to crimes, numerous studies have suggested that the expansion of night-time economy can creat urban spatial hot spots of

reported anti-social behaviour and violence (e.g., Bromley and Nelson, 2002; Jackson and Owens, 2011; van Liempt et al., 2014; Philpot et al., 2019). In particular, Jackson and Owens (2011) found that minor crimes rose in the areas where bars were near transit stops that had new late-night services, which could be possibly due to more night-time activities. The implementation of the Night Tube will undoubtedly lead to growing night-time journeys and activities, but whether this can result in a higher crime rate remains underexplored. In this study, London's LSOA level crime data provided by the Metropolitan Police Service is used for our causal modeling to answer the question (London Datastore, 2021). Crime rate per thousand population is calculated. A limitation of this dataset should be noted. Crime data is not reported with occurring time of day (we cannot confine the crime data to weekend night as the road crash data). Nonetheless, the Metropolitan Police Service has published a dataset of public transport crimes at London level, which can be used for a supplementary analysis.

We have further included three sets of covariates in our model to adjust for factors correlated to the outcomes of interest and proximity to the Night Tube stations (i.e., treatment). First, baseline socio-demographic characteristics (e.g., population and employment characteristics, dwelling types, cars or vans in the area, and deprivation) are included as they are closely related to the outcomes of interest. Note that the percentage of different dwelling types (e.g., detached, terraced, flat) is included in our house price models. The measure of LSOA level deprivation used in this study is the index of multiple deprivation (IMD), which is an overall measure combining a set of domains of deprivation (e.g., income, education, health, living environment; please refer to Gov.UK, 2020 for more details). A more deprived area has a higher IMD. Moreover, due to the unavailability of the data for traffic exposure at LSOA level, population and employment data can be used as an alternative for zone-level road crash modeling (e.g., Graham and Glaister, 2003; Li et al., 2015). The second set is land use data (Office of the Deputy Prime Minister, 2006), including the percentage of domestic buildings (e.g., dwellings), non-domestic buildings (e.g., shops, offices), and greenspace (e.g., parks). In addition, road network characteristics are crucial for road crash modeling (Noland and Quddus, 2004). Therefore, we have collected the data containing the density (km/km^2) of three classes of road networks, class A road, class B road, and minor road (the three road classes in the UK are defined in the *Design Manual for Roads and Bridges*, please refer to Department for Transport, 2016 for more details) for each LSOA. Furthermore, a set of dummy variables for London boroughs are also included to account for unobserved heterogeneity across boroughs.

Another critical issue is concerned with other interventions that could affect the outcomes of interest, and were implemented differently in the treated and control areas. In particular, the London congestion charge (LCC), which was introduced alongside a series of other changes to the transport system (Givoni, 2011), has been found to be able to reduce road crashes and casualties in previous studies (e.g., Li et al., 2012; Green et al., 2016). Also, road pricing interventions can also affect house prices (e.g., Percoco, 2014; Tang, 2021). Nonetheless, LCC was initially introduced in 2003, and some major changes were imposed during the period from 2005 to 2010, much earlier than our study period. Therefore, it is reasonable to assume that the impacts of LCC would not substantially affect our estimation results. Concerning the crime rate, the British Transport Police patrol the Night Tube stations to guarantee safe travels (Visit London, 2021), which could more or less control the crime rate in our treated areas. Thus, if the implementation of the Night Tube is

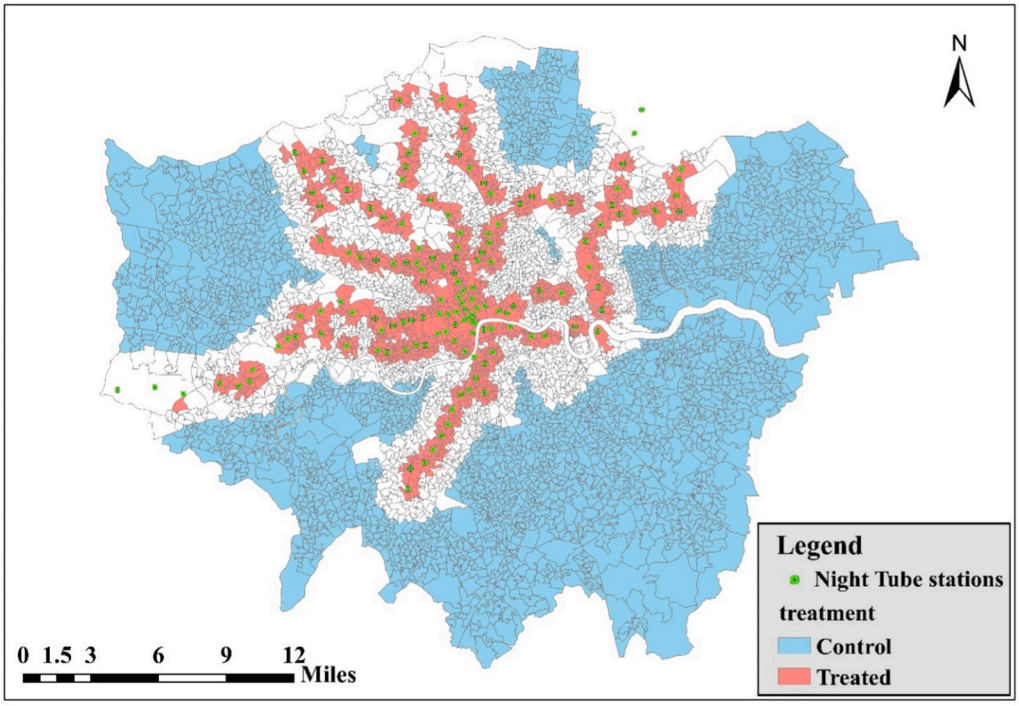


Fig. 2. Selected treated and control LSOAs.

found to increase the crime rate, such impacts may be underestimated.

3.2. Selection of treated and control groups

The Night Tube stations are matched with LSOAs and MSOAs based on their geographical proximity. Specifically, we create a circular buffer with a radius of 800 m for each Night Tube station. If the centroid of an LSOA is located within the buffer area, it is selected into our treated group. Note that the LSOAs near the Night Overground stations are excluded from the dataset. For the control group, an important property it should have is the independence of the treatment. That is, the control group should not have received the treatment either directly or indirectly through proximity to or interaction with the treated group (Li et al., 2012). Although the maximum affected area of the Tube station is unclear, previous findings have indicated that the actual walking and cycling distance of transfer trips to and from the urban rail transit stations mostly ranges within 3 km (e.g., Millward et al., 2013; Zhao and Li, 2017; Shelat et al., 2018; Paydar et al., 2020; van Mil et al., 2021). Therefore, the LSOAs that do not intersect with the 2-km circular buffers of the Night Tube stations are selected into our control group to ensure that they were not or at least limitedly affected by the Night Tube. The selected treated and control LSOAs are shown in Fig. 2. Furthermore, to explore spatial variations in the impacts of the Night Tube, we specify a separate model for the inner London area for each outcome of interest. Table 2 summarizes the descriptive statistics of our outcomes and covariates.

The pre- and post-treatment periods are determined by the launch year of the Night Tube (i.e., 2016) and data availability for each outcome. Specifically, London's night-time economy dataset is available up to 2017, and therefore we use 2015 as our pre-treatment period, and 2017 as our post-treatment period. Moreover, we use two-year (i.e., 2017 – 2018) and three-year (i.e., 2017 – 2019) post-treatment observations for crimes and road crashes respectively because of the following two reasons. First, the LSOA level crime data for the first half of 2019 is not provided online. Second, the Night Tube service has been suspended since March 2020 due to the COVID-19 pandemic. For each outcome, an equal-length pre-treatment period is used, and the data for 2016 is not included. Furthermore, considering that the house price might respond to the official announcement of the new service prior to its formal operation (e.g., Surrenden Invest, 2016; such effect is termed as *anticipated capitalization effect* in Agostini and Palmucci, 2008), we use year 2014 as the time point of treatment when quantifying the response of house prices to the announcement of the Night Tube with 2012 – 2013 being the pre-treatment period, and 2014 – 2015 being the post-treatment period. The impact after the launch of the Night Tube is also estimated using 2017 – 2018 as the post-treatment period. Hereafter, we term the periods after the announcement and the launch of the Night Tube as post-announcement and post-launch periods, respectively.

4. Methodology

The analysis is carried out on a dataset with N spatial units namely LSOAs or MSOAs indexed $i = 1, \ldots, N$, and two periods $t = 0, 1$, representing pre- and post-treatment periods. Let Y_{it} be the observed outcome for unit i at time t, and D_{it} be the treatment indicator at time t. We have $D_{i0} = 0$ for all units (i.e., all units are untreated in the pre-treatment period), and thus we can reduce notation by writing $D_i = D_{i1}$. In our case, $D_i = 1$ if the centroid of unit i is located within an 800-m buffer of the Night Tube stations, and $D_i = 0$ if unit i does not intersect with any 2-km buffer of the stations. Using the potential outcome notation of Rubin (1974), let $Y_{it}(1)$ and $Y_{it}(0)$ be the potential outcomes for unit i at time t if it receives treatment and does not receive treatment, respectively. The fundamental problem of causal inference is that we cannot observe both $Y_{it}(1)$ and $Y_{it}(0)$ for the same unit. The observed outcome in the post-

Table 2
Summary of baseline outcome variables and covariates.

Variables	Treated group	Control group
Outcomes		
Median house prices (GBP)	521276.2 (410419.5)	291457.2 (135616.5)
Number of night-time workplaces	190.8 (351.4)	73.66 (44.51)
Crash rate (per thousand population)	0.524 (1.281)	0.227 (0.305)
Crime rate (per thousand population)	93.56 (179.0)	48.04 (44.47)
Covariates		
Population density (per km^2)	13336.5 (7666.2)	7440.0 (4552.7)
Employment density (per km^2)	6231.1 (3883.0)	3163.5 (1868.8)
IMD	24.67 (11.83)	21.10 (12.16)
Cars or vans in the area	432.8 (178.2)	647.8 (185.9)
Percentage of detached house or bungalow	0.036 (0.056)	0.096 (0.132)
Percentage of semi-detached house or bungalow	0.112 (0.152)	0.278 (0.200)
Percentage of terraced house or bungalow	0.175 (0.159)	0.286 (0.196)
Percentage of flat, maisonette, or apartment	0.676 (0.258)	0.339 (0.247)
Percentage of domestic buildings	0.156 (0.072)	0.122 (0.055)
Percentage of non-domestic buildings	0.105 (0.094)	0.041 (0.044)
Percentage of greenspace	0.147 (0.132)	0.236 (0.202)
Class A road density (km/km^2)	2.124 (2.283)	0.992 (1.316)
Class B road density (km/km^2)	0.589 (1.235)	0.311 (0.790)
Minor road density (km/km^2)	11.09 (4.674)	8.799 (3.894)

Note: Mean values are reported with standard deviations in parenthesis.

treatment period is $Y_{i1} = D_i Y_{i1}(1) + (1 - D_i) Y_{i1}(0)$. In other words, we observe only $Y_{i1}(1)$ for the treated units, and $Y_{i1}(0)$ for the control ones. The primary objective is to estimate the average treatment effect on the treated (ATT):

$$\theta = E[Y_{i1}(1) - Y_{i1}(0) | D_i = 1] \tag{1}$$

A before-and-after comparison for treated units may be biased due to the temporal trend in outcomes, and simply taking the difference in post-treatment outcomes between treated and control groups may be subject to problems of confounding (Anupriya et al., 2020). The DID estimation method is adopted in this study, which has been a widespread approach for causal inference since the work by Ashenfelter and Card (1985). DID is able to account for time-invariant unobserved confounding and temporal trend by using information for both treated and control groups in both pre- and post-treatment periods (Graham, 2021). A great number of recently published studies on policy evaluations for transport systems have used DID based approaches (e.g., Alemi et al., 2018; Li et al., 2019; Wang and Lindsey, 2019; Anupriya et al., 2020; Kuo et al., 2021; Zheng et al., 2021).

In this study, we use a double/debiased machine learning based difference-in-differences approach to make causal inference on the social impacts of the London Night Tube. DMLDID is a modified version proposed by Chang (2020), which combines Abadie's semiparametric DID estimator and double/debiased machine learning. It has higher functional flexibility and allows researchers to handle a rich set of control variables while taking strengths of the DID research design. The remainder of this section discusses the specification of the traditional DID model, and provides theoretical details on DMLDID.

4.1. Traditional DID model

The traditional linear DID model without covariates can be written as:

$$Y_{it} = \mu + \alpha \bullet D_i + \delta \bullet t + \tau \bullet D_{it} + \varepsilon_{it} \tag{2}$$

where μ, α, δ, and τ are the parameters to be estimated, and ε_{it} is an error term with mean zero. The treatment effect is captured by τ.

A critical assumption of DID is termed *parallel trend assumption*, which states that the average outcomes of treated and control groups would have followed parallel trends over time in the absence of the treatment:

$$E[Y_{i1}(0) - Y_{i0}(0) | D_i = 1] = E[Y_{i1}(0) - Y_{i0}(0) | D_i = 0] \tag{3}$$

If the assumption is violated, the DID estimator can be biased. In practice, a vector of covariates can be added to the model to partly account for this issue. However, the traditional model performs poorly when dealing with high-dimensional confounding and treatment effect heterogeneity. Another common issue of the traditional DID is model misspecification. To address these issues, Chang (2020) proposed a DMLDID estimator that combines Abadie's semiparametric DID estimator (Abadie, 2005) and double/debiased machine learning (Chernozhukov et al., 2018).

4.2. Double/debiased machine learning based DID estimator

Abadie (2005) identify the ATT under the following two assumptions in *repeated outcomes* cases (i.e., we observe $\{Y_{i0}, Y_{i1}, D_i, X_i\}$; we can reduce notation by dropping unit subscript i $\{Y_0, Y_1, D, X\}$). The first assumption is termed *conditional parallel trend assumption*, which states that conditional on a set of covariates X_i, the average outcomes of treated and control groups would have followed parallel trends over time in the absence of the treatment:

$$E[Y_1(0) - Y_0(0) | X, D = 1] = E[Y_1(0) - Y_0(0) | X, D = 0] \tag{4}$$

Second, the support of propensity scores (i.e., conditional probability of receiving the treatment) for the treated units is a subset of the support for the control ones:

$$P(D = 1) \rangle 0 \text{ and with probability one } P(D = 1|X) \langle 1 \tag{5}$$

The ATT is identified as:

$$\theta_0 = E\left[\frac{Y_1 - Y_0}{P(D = 1)} \bullet \frac{D - P(D = 1|X)}{1 - P(D = 1|X)}\right] \tag{6}$$

We can estimate the ATT as:

$$\widehat{\theta} = \frac{1}{N} \sum\nolimits_{i=1}^{N} \frac{Y_{i1} - Y_{i0}}{\widehat{p}} \bullet \frac{D_i - \widehat{g}(X_i)}{1 - \widehat{g}(X_i)} \tag{7}$$

where $\widehat{p}$ is the estimator of $p_0 = P(D_i = 1)$, $\widehat{g}(X_i)$ is the estimator of the propensity score $g_0(X_i) = P(D_i = 1|X_i)$. Abadie's semiparametric DID estimator is $\sqrt{N}$-consistent and asymptotically normal when the propensity score is estimated by conventional nonparametric methods. However, when $\widehat{g}(X_i)$ is a machine learning (ML) estimator, $\widehat{\theta}$ is not necessarily $\sqrt{N}$-consistent because the score function based on Eq. (6) has a non-zero directional derivative with respect to the propensity score, and moreover, ML estimators usually converge at a rate slower than $N^{-\frac{1}{2}}$ due to the regularization bias (Chang, 2020). To address these problems, a DMLDID estimator has been proposed based on the cross-fitting algorithm and the Neyman-orthogonal score of Chernozhukov et al. (2018):

$$\varphi = \frac{Y_1 - Y_0}{P(D=1)} \bullet \frac{D - P(D=1|X)}{1 - P(D=1|X)} - \theta_0 - c \tag{8}$$

where $c = \frac{D-P(D=1|X)}{P(D=1)\bullet(1-P(D=1|X))} E[Y_1 - Y_0|X, D=0]$ is an adjustment term. In this approach, the use of Neyman-orthogonal scores allows consistent estimation of the parameter of interest, and cross fitting removes the bias due to overfitting. For more details, please refer to Chernozhukov et al. (2018). Specifically, the procedures for using DMLDID to estimate the ATT can be stated as follows (Chang, 2020):

Step 1. Form K random partitions for the data sample S by K-fold splitting. For the sake of simplicity, we can use the same size $n = \frac{N}{K}$ for each fold S_k. Define an auxiliary subset $S_k^c = S \backslash S_k$ for each fold.

Step 2. Use S_k^c to construct the estimators for p_0, g_0 and l_0; $l_0 = E[Y_1 - Y_0|X, D=0]$. Specifically, $\widehat{p}_k = \frac{1}{n}\sum_{i \in S_k^c} D_i$, and $\widehat{g}_k$, $\widehat{l}_k$ are constructed using ML methods that converge at a rate faster than $N^{-\frac{1}{4}}$ (e.g., Lasso or logit Lasso, Random forests, Support vector machines). Individual observations with extreme propensity scores are discarded in practice. In our modeling procedures, the random forests algorithm (R package **randomForest**) is used to estimate the nuisance parameter l_0, as it can well handle different types of variables and model complex nonlinear relationships.

Step 3. Construct the intermediate debiased ATT estimator using the remaining data in fold S_k:

$$\widetilde{\theta}_k = \frac{1}{n}\sum_{i \in S_k} \frac{D_i - \widehat{g}_k(X_i)}{\widehat{p}_k(1 - \widehat{g}_k(X_i))} \bullet (Y_{i1} - Y_{i0} - \widehat{l}_k(X_i)) \tag{9}$$

Step 4. Aggregate the estimators, $\widetilde{\theta}_{DML} = \frac{1}{K}\sum_{k=1}^{K} \widetilde{\theta}_k$.

The DMLDID estimator has been proved to be able to achieve $\sqrt{N}$-consistency and asymptotic normality by using the ML methods that converge at a rate faster than $N^{-\frac{1}{4}}$, and a consistent estimator for the asymptotic variance of $\widetilde{\theta}_{DML}$ has been provided for associated statistical inference.

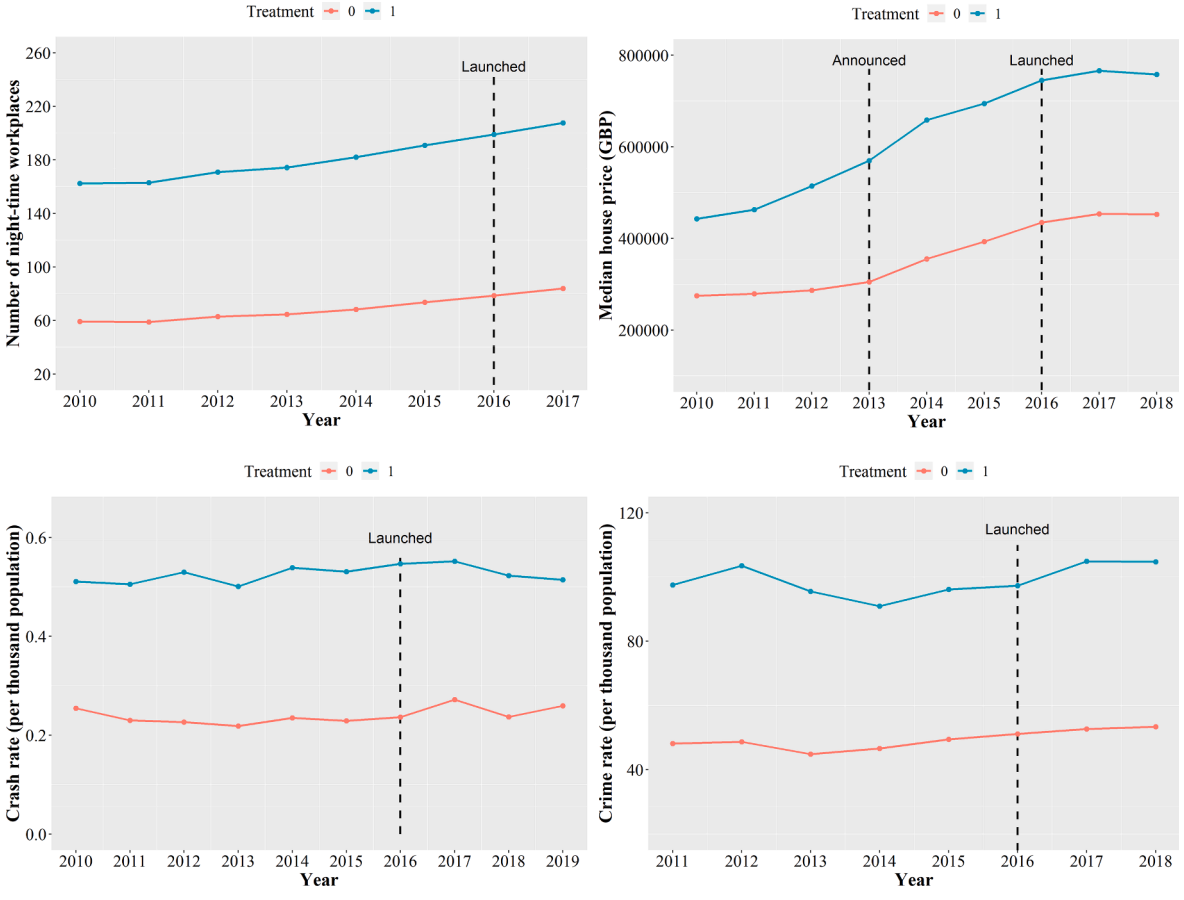


Fig. 3. Temporal trends in average outcomes for the treated and control groups.

5. Results and discussion

5.1. Preliminary analysis

A series of preliminary analysis on our outcomes is presented in this section. First, as in Hastings (2004), the parallel trend assumption is partly realizable when observations in multiple pre-treatment years are available. That is, the information of outcomes in periods prior to the treatment assignment can be used to check the validity of the assumption. Therefore, we follow Hastings (2004) to conduct a graphical test that uses more observations in pre-treatment years to assess whether the control group is suitable to mimic the temporal trend of the treated group (see Fig. 3). The observed temporal trends in our outcomes for the two groups are similar but not totally parallel.

We further present a difference-in-means analysis as in Ait Bihi Ouali et al. (2021). The spatial distribution of the changes between pre- and post-treatment period in our four outcomes is first illustrated in Fig. 4. It is clear that the house price in areas along the Night Tube routes experienced a huge pre- to post-treatment increase. The change in crime rate showed a similar pattern. For the other two outcomes, few intuitive patterns can be drawn from the figures. Fig. 5 presents the average change between the pre- and post-treatment periods for the treated and control groups, respectively, and we further differentiate the two to compute the difference-in-means (see the Diff. column in Table 3). We find that the Night Tube might have significant causal impacts on the outcomes of interest. However, we note that the difference-in-means results should be only considered as preliminary because the observed difference may be subject to confounding. This issue can be partly addressed by employing a full DID analysis including relevant covariates.

5.2. DMLDID estimates

The impacts of the Night Tube on our outcomes are formally estimated with DMLDID. Tables 4 and 5 report the DMLDID estimates for Greater London (i.e., overall) and inner London, respectively.

Our results indicate that the post-launch impacts on the number of night-time workplaces are positive. In particular, the cultural and leisure activities related workplaces showed a great response with an increase of 3.07 (3.11 %), and the figure is 5.28 (4.36 %) in the inner London area. According to a bus user survey conducted by TfL in 2014, compared to day times, journeys with leisure purposes share a much higher percentage at night (Transport for London, 2014b; Mayor of London, 2018). Therefore, the growth in the number of night-time cultural and leisure sectors could be attributable to the increased night-time leisure activities promoted by the Night

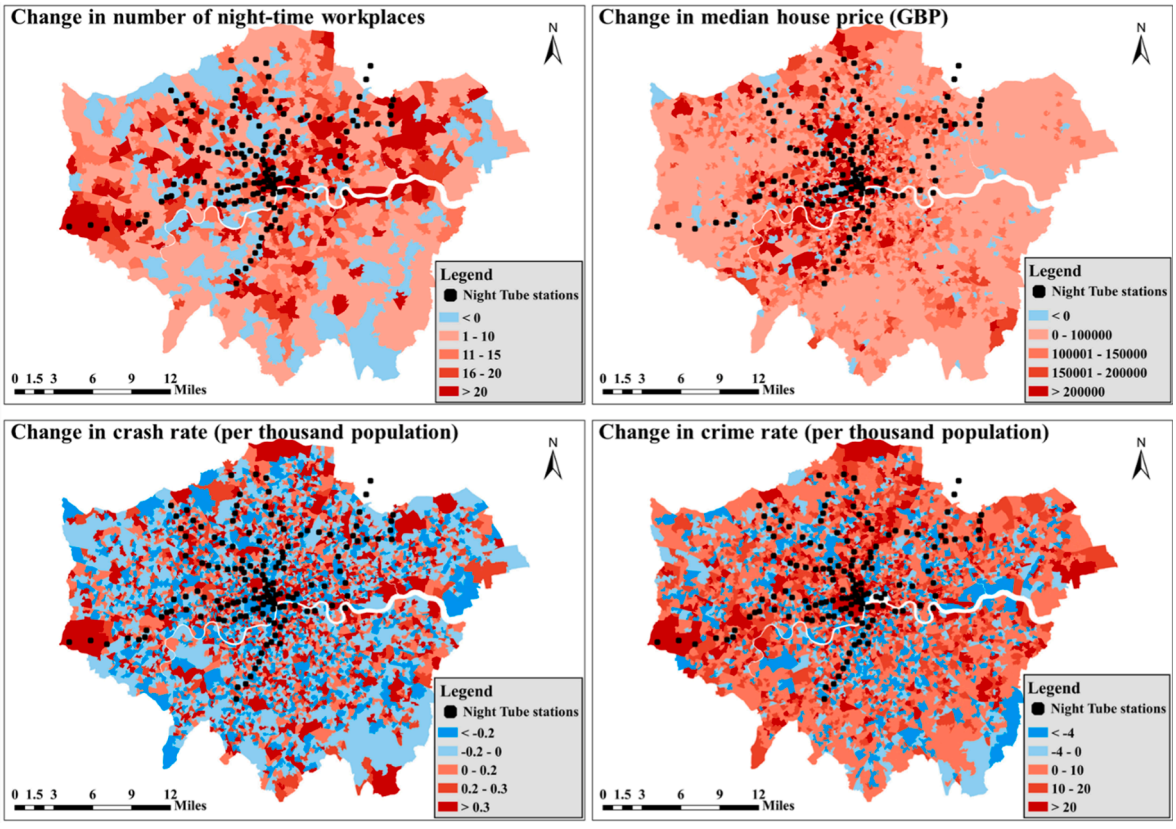


Fig. 4. Spatial distribution of the changes in outcomes between pre- and post-treatment periods.

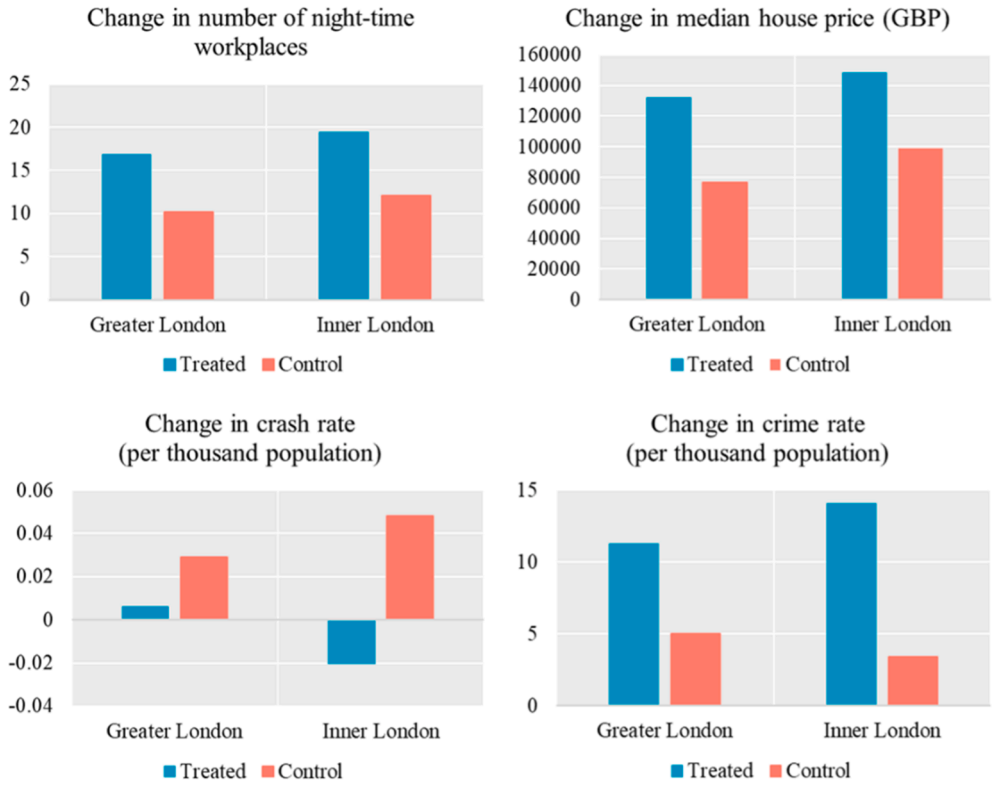


Fig. 5. Average change between pre- and post-treatment periods for the treated and control groups.

Table 3
Summary of difference-in-means estimates.

Outcomes	Overall			Inner London		
	Treated	Control	Diff.	Treated	Control	Diff.
Median house prices (GBP)	132103.29	76755.29	55347.80	147931.97	99219.79	48712.18
Number of night-time workplaces	16.849	10.220	6.629	19.404	12.109	7.295
Crash rate (per thousand population)	0.006	0.029	−0.023	−0.020	0.048	−0.068
Crime rate (per thousand population)	11.286	5.015	6.271	14.055	3.428	10.627

Tube. However, the impacts on the other two categories of night-time workplaces are insignificant. Nonetheless, such results do not totally mean that the implementation of the Night Tube had little impact on the night-time economy. It should be noted that due to data restriction, the night-time activities data for small geographic areas such as LSOAs is not available, which makes it impossible for us to directly quantify the impacts on night-time activities. Moreover, the data reported in Mayor of London (2018) indicate that during the period from 2001 to 2017, the pub numbers experienced a notable reduction, whereas the pub-related employment increased, which was due to the fact that the reduction in the number of small pubs with fewer than 10 staff was much larger than the rise in the number of larger pubs. Thus, the impacts on the night-time economy are recommended to be re-analyzed by using more detailed data for night-time activities and employment.

The median house price in areas served by the Night Tube is found to additionally increase by £36,275 (5.88 %) in the post-announcement period. In the inner London area, the impact is larger with a value of £44,383 (6.33 %). By controlling for the factors that could affect the dynamics in house prices as discussed in section 3.1, DMLDID provides much smaller estimates than those reported in difference-in-means analysis (see Table 3). The post-launch effect is 7.56 % overall, and 9.47 % in the inner London area, which can be interpreted as the cumulative impacts of the Night Tube service since its announcement. To be noted is the fact that the Brexit vote (i.e., vote to leave the European Union) has affected London's house price greatly since 2016 due to the resulting uncertainty (Standard UK, 2021). Therefore, the post-launch impacts should be treated cautiously. The rise in house prices associated with the Night Tube service can be largely explained by the higher accessibility to public transport services at night and the expansion of the local night-time economy (e.g., Olt et al., 2019; Pinke-Sziva et al., 2019; Smith et al., 2019).

We now assess the Night Tube from the perspective of safety. Regarding the impacts on road crashes, the causal link between the implementation of the Night Tube service and the reduction in road crashes is intuitive. More people would choose to travel by the

Table 4
Summary of DMLDID estimates.

Outcome and evaluation periods	Effect	Percentage	Std. Err.
Night-time workplace (pre-treatment: 2015, post-treatment: 2017)			
1 – cultural and leisure activities	3.069	3.11 %	1.924
2 – activities which support night time cultural and leisure activities	0.142	0.72 %	0.869
3 – activities which support wider social and economic activities	0.684	1.01 %	1.519
House price (pre-treatment: 2012–2013, post-announcement: 2014–2015, post-launch: 2017–2018)			
Median house price (GBP): post-announcement	36275***	5.88 %	6596
Median house price (GBP): post-launch	52935***	7.56 %	7115
Road crash rate (pre-treatment: 2013–2015, post-treatment, 2017–2019)			
Weekend night-time crash rate (800-m buffer)	−0.013	−2.39 %	0.032
Weekend night-time crash rate (400-m buffer)	−0.055	−7.00 %	0.045
Weekend night-time casualty rate (400-m buffer)	−0.096*	−9.91 %	0.055
Crime rate (pre-treatment: 2014–2015, post-treatment: 2017–2018)			
Criminal damage to motor vehicles	0.097	3.71 %	0.128
Robbery of personal property	0.841***	13.20 %	0.184
Violence against the person	1.824***	5.60 %	0.613
Motor vehicle-related theft and handling	−0.401	−3.23 %	0.298
Other theft (e.g., theft person, theft of pedal cycles)	5.160***	9.66 %	1.250
Other notifiable offences (e.g., going equipped, other)	−0.019	−0.91 %	0.094

Note: Significance levels are denoted as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 5
Summary of DMLDID estimates for the inner London area.

Outcome and evaluation periods	Effect	Percentage	Std. Err.
Night-time workplace (pre-treatment: 2015, post-treatment: 2017)			
1 – cultural and leisure activities	5.279**	4.36 %	2.284
2 – activities which support night time cultural and leisure activities	0.454	2.24 %	0.914
3 – activities which support wider social and economic activities	−0.103	−0.13 %	1.721
House price (pre-treatment: 2012–2013, post-announcement: 2014–2015, post-launch: 2017–2018)			
Median house price (GBP): post-announcement	44383***	6.33 %	6977
Median house price (GBP): post-launch	73733***	9.47 %	8151
Road crash rate (pre-treatment: 2013–2015, post-treatment: 2017–2019)			
Weekend night-time crash rate (800-m buffer)	−0.039	−5.86 %	0.035
Weekend night-time crash rate (400-m buffer)	−0.098*	−10.10 %	0.041
Weekend night-time casualty rate (400-m buffer)	−0.149***	−12.61 %	0.053
Crime rate (pre-treatment: 2014–2015, post-treatment: 2017–2018)			
Criminal damage to motor vehicles	0.152	5.89 %	0.145
Robbery of personal property	1.119***	19.10 %	0.219
Violence against the person	2.785***	7.61 %	0.802
Motor vehicle-related theft and handling	−1.137***	−8.32 %	0.341
Other theft (e.g., theft person, theft of pedal cycles)	7.950***	14.39 %	1.571
Other notifiable offences (e.g., going equipped, other)	−0.025	−1.04 %	0.113

Note: Significance levels are denoted as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Tube leaving their cars at home. However, the post-launch causal impacts are found to be insignificant with a value of −0.013 (-2.39 %). Motivated by Jackson and Owens (2011) who suggested that the reduction in drunk driving induced by night-time transport services are closely related to the distance to metro stations, we have further tried a smaller buffer with a radius of 400 m to select the treated group and performed a supplementary analysis. Results indicate that the crash reduction effect is much larger in areas closer to the Night Tube stations, implying that the transport modal shift associated with the Night Tube could be relevant to the proximity to the stations. We have also estimated the impact on crash-related casualties. Results show that the implementation of the Night Tube caused a significant reduction of 0.096 (9.91 %) in casualty rate, indicating that the service had created a safer environment of road traffic for night-time travelers. Similar impact is found in the inner London area.

The post-launch impacts on crime rates are found to be positive and significant for the following three categories, robbery of personal property (0.84, 13.20 %), violence against the person (1.82, 5.60 %), and other theft (5.16, 9.66 %). Moreover, the impacts are larger in the inner London area. Similar results were obtained by Jackson and Owens (2011). They found that the expansion of metro service hours in Washington DC caused an increase in arrests for minor crimes, for which a reasonable explanation is the increase in late-night activities. As a contrast, the rate of motor vehicles related crimes (e.g., criminal damage and theft) showed limited response to the implementation of the Night Tube. Interestingly in the inner London area, motor vehicle-related theft and handling were decreased. A supplementary analysis is conducted using the public transport crime data published by the Metropolitan Police Service. Temporal trends in the crime rate on the London Tube, Overground, and bus networks are presented in Fig. 6. It can be seen

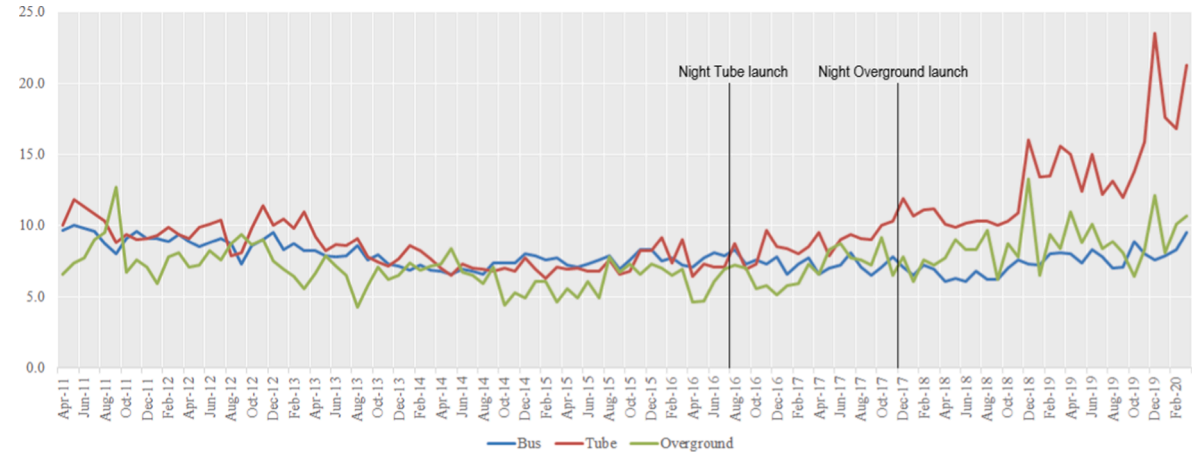


Fig. 6. Temporal trends in crime rates (per million passenger journeys) on the London Tube, Overground, and bus networks.

that the crime rate on the Tube had been similar to that on the bus networks until 2017. Subsequently, it experienced a notable increase during the period from 2017 to 2019, and was much higher than that on bus networks. Similar temporal trend was observed for the crime rate on the London Overground (before and after the launch of the Night Overground service in December 2017). Such changes could partly be attributable to the night-time services on both systems.

5.3. Robustness test

We now perform a robustness test by examining the existence of a treatment effect on our outcomes using the pre-treatment observations only. For instance, a pre-announcement period from 2012 to 2013 is used as our pseudo post-treatment period for the house price. That is, we estimate the treatment effect of the Night Tube on house prices before its announcement, and for simplicity, we term such effect as pseudo effect hereafter. Because the treatment had not been assigned (i.e., the Night Tube service had not been announced until the end of 2013), the model should report a null or insignificant pseudo effect estimate. For the other three groups of outcomes, the corresponding pre-launch periods defined previously are regarded as post-treatment periods in this part, and an equal-length pre-treatment period is used.

Table 6 reports the pseudo effect estimates. All the pseudo effects are insignificant except that for a crime rate outcome, namely other theft. In section 5.2, a significant increase associated with the Night Tube was found for other theft in the post-launch period. However, the result here indicates that the previous estimate is unreliable as the parallel trend assumption is violated. We note that such a significant pseudo effect could be due to the unobserved confounding issue. In this case, the existence of unobserved confounders that can affect the rate of other theft play a critical role. Therefore, the estimated post-launch increase in other theft cannot be solely attributable to the Night Tube. As a contrast, the pseudo effects are insignificant for the other two crime rate outcomes that were found to be significantly affected by the Night Tube service (i.e., robbery of personal property and violence against the person), assuring the robustness of corresponding post-launch effects. Lastly, the pseudo effects for the number of cultural and leisure activities related workplaces, house prices, and our road safety outcomes are statistically insignificant, indicating that the inherent difference in outcome dynamics between the treated and control areas has been largely corrected by controlling for the observed confounders, and therefore the estimated post-launch effects on these outcomes are also robust.

5.4. Sensitivity analysis for buffer size

Furthermore, a sensitivity analysis is conducted for the buffer size used to selecting our control groups. A different buffer size can produce a different control sample. Specifically, we use 2.2 km and 2.5 km as alternatives and re-analyzed the impacts on our four outcomes. Note that we re-analyze the outcomes that showed the greatest response to the Night Tube service in each outcome group. The results are shown in Table 7, suggesting that the causal impact estimate is not too sensitive to the buffer size. Yet the absolute values of the estimated impacts exhibit a systematic increase when increasing the buffer size. A major reason is that increasing the buffer size would discard more potential control LSOAs, and thereby make the selected sample more difficult to satisfy the model assumptions (please refer to section 4.2). Specifically, for some treated LSOAs, especially those located in the inner area, it is more difficult to find sufficient similar controls for them when increasing the buffer size. As a result, the DMLDID estimator tends to overestimate the causal impacts as these treated areas usually exhibit larger inherent differences with others (e.g., higher house prices, more night-time industries). In this case, a reasonable buffer size should ensure that the control areas were not affected by the Night Tube service as previously discussed in section 3.2, and a considerable sample size of the control group can be guaranteed. Therefore, a buffer size of 2 km is the most proper one.

Table 6
Summary of pseudo effect estimates.

Outcome and pseudo evaluation periods	Effect	Std. Err.
Night-time workplace (pre-treatment: 2014, post-treatment: 2015)		
1 – cultural and leisure activities	0.437	1.131
2 – activities which support night time cultural and leisure activities	0.556	0.785
3 – activities which support wider social and economic activities	1.462	1.090
House price (pre-treatment: 2010–2011, post-treatment: 2012–2013)		
Median house price (GBP)	26,028	24,643
Road crash rate (pre-treatment: 2010–2012, post-treatment: 2013–2015)		
Weekend night-time crash rate (800-m buffer)	0.012	0.022
Weekend night-time crash rate (400-m buffer)	0.030	0.031
Weekend night-time casualty rate (400-m buffer)	0.056	0.041
Crime rate (pre-treatment: 2012–2013, post-treatment: 2014–2015)		
Criminal damage to motor vehicles	0.057	0.120
Robbery of personal property	−0.104	0.192
Violence against the person	0.831	0.600
Motor vehicle-related theft and handling	0.403	0.295
Other theft (e.g., theft person, theft of pedal cycles)	−5.949***	1.644
Other notifiable offences (e.g., going equipped, other)	0.018	0.101

Note: Significance levels are denoted as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 7
Sensitivity analysis for buffer size.

Outcome	Buffer size	N_{treat}	$N_{control}$	Effect	Std. Err.
House price	2.0 km	961	2086	36275***	6596
(median house price in GBP)	2.2 km	961	2000	38524***	6254
	2.5 km	961	1875	41964***	6525
Night-time workplace	2.0 km	219	409	3.069	1.924
(cultural and leisure activities)	2.2 km	219	392	3.299*	1.995
	2.5 km	219	370	3.321*	2.003
Road crash rate	2.0 km	357	2225	−0.096*	0.055
(casualty rate within 400-m buffer)	2.2 km	357	2129	−0.100*	0.057
	2.5 km	357	1990	−0.109*	0.057
Crime rate	2.0 km	1036	2215	0.841***	0.184
(robbery of personal property)	2.2 km	1036	2119	0.892***	0.187
	2.5 km	1036	1980	0.942***	0.207

Notes: Significance levels are denoted as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. N_{treat} and $N_{control}$ are the sample sizes of the treated and control groups, respectively.

6. Conclusions

The worldwide growth of the night-time economy has led to an increasing attention on urban night-time public transport services. Transport for London launched the Night Tube service in 2016 to meet the constantly growing night-time transport demand and support London's night-time economy. Yet limited ex-post evaluation has been conducted for the London Night Tube. This study presents new empirical evidence on the causal impacts of the Night Tube on relevant social outcomes, covering economic and safety dimensions. A double/debiased machine learning based difference-in-differences approach is employed to make causal inference, which combines the strengths of machine learning algorithms and the DID research design. Compared to the traditional DID models, DMLDID has higher functional flexibility particularly when handling high-dimensional confounding and treatment effect heterogeneity, which can be used more widely for monitoring the causal impacts of transport interventions.

Our analysis focuses on the following four outcomes, namely night-time economy, house prices, road crashes and related casualties, and crimes. Results indicate a positive impact of the Night Tube on cultural and leisure activities related businesses, which is possibly due to increasing night-time activities. Also, the announcement and the implementation of the Night Tube led to a rise in the house price. We have further evaluated the Night Tube service from the perspective of safety. Although an insignificant reduction in the crash rate associated with the Night Tube is observed, the impact is found to be larger in areas closer to the Tube stations. Moreover, the impact on crash-related casualties is significant with a value of −0.096 (-9.91 %), indicating that the Night Tube has created a safer road traffic environment. Regarding the crime rate, robbery of personal property and violence against the person in areas served by the Night Tube experienced a notable increase of 13.2 % and 5.6 %, respectively. A supplementary analysis also suggests that the crime rate on the London Tube and Overground has risen significantly since the launch of their night-time services. With these findings, it is worth summarizing some policy implications.

First, the increase in house price is inevitable during the expansion of the night-time economy. For London, one of the most

expensive cities in the world but with many low-income households, whether the rise in house prices has induced gentrification remains underexplored. Here we further provide some preliminary results based on the raw data. Specifically, the rise in house prices in the most deprived areas with IMD ranging from 40 to 65 not served by the Night Tube is £120,000 during the study period. As a contrast, house prices in areas with similar baseline IMD but served by the Night Tube increased by £180,000, which could be a sign of increased gentrification. As suggested by Kwan (2020), an important force for resisting the trend of gentrification is to integrate new businesses into the local culture. As a counter-example, a new night-time business opened in the outer London, which is largely the same as those in the city center, may erode local characteristics.

Second, a great number of studies have suggested that a cheap, fast and highly accessible public transport service can encourage more people to leave their cars at home, hence reduce the road crash and casualty rates (e.g., Scagnolari et al., 2015; Lichtman-Sadot, 2019). Our results provide some new evidence on its safety effects at night. Therefore, urban planners and policy makers are suggested to take a prior consideration of the public transport service when seeking to improve the night-time mobility.

Lastly, safety is of central importance for night-time cities, as said by Mayor of London Sadiq Khan 'the safety and wellbeing of London's residents, workers, and visitors is my top priority' (Mayor of London, 2017; 2018). The British Transport Police provides more than 100 officers to patrol the Night Tube stations to guarantee safe travel at all time (Visit London, 2021). Yet our results indicate that in areas served by the Night Tube, the crime rate experienced a notable increase, which was possibly due to more night-time activities. Also, data provided by the Metropolitan Police Service suggests that the crime rate on the London Tube has substantialy increased since the launch of its night-time service. Therefore, more efforts should be made to keep the Night Tube service and surrounding neighborhoods safe during the expansion of the night-time economy.

Some limitations of this study should be noted. First, only the short-term impacts of the Night Tube service have been evaluated for house prices and the night-time economy. The long-term impacts and resulting consequences such as increased gentrification should be monitored over time. Second, data for night-time employment, night-time mobilities and activities for small geographic areas are not publicly available, and therefore the impacts on the night-time economy cannot be accurately estimated. Future research using more detailed and comprehensive dataset is highly recommended, which can help better reveal the response of the night-time economy to the addition of a new night-time transport service. Lastly, the current version of DMLDID is particularly suitable for a standard two periods, two groups setup. An extension that allows multiple time periods setup would be helpful for handling more complex cases (e.g., time-varying treatment and time-varying effects; see for instance Callaway and Sant'Anna, 2021).

CRediT authorship contribution statement

Yingheng Zhang: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft. **Haojie Li:** Conceptualization, Data curation, Supervision, Writing – review & editing. **Gang Ren:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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